

Design, Development and Experimentation of an LLM Agent for the Workplace Safety

A Hybrid GraphRAG Approach for Institutional OSH Compliance

Master's Degree in Statistical Methods and Applications

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SAPIENZA
UNIVERSITÀ DI ROMA



Occupational Safety: High Stakes Knowledge Domain

1 Introduction

THE CHALLENGE

- Thousands of documents across **INAIL**, **EU-OSHA** and national laws
- Unstructured PDFs, nested tables, complex legal citations
- Errors → **legal liability** + **safety risks**

Why LLMs?

Natural Language

Query regulations in plain language

Synthesis

Aggregating multi-source information

Scale

Process beyond human capacity

🎯 **RES OBJ:** *Designing grounded OSH agents via authoritative sources & validity aware legal data*



The Reliability Gap: Why Standard LLMs Fail

1 Introduction

Generative Approach (Pure LLM)



- **Fluent Hallucinations**
- Models generate plausible-sounding but potentially false answers
- Repealed laws cited as valid

Hybrid GraphRAG Approach (Proposed)

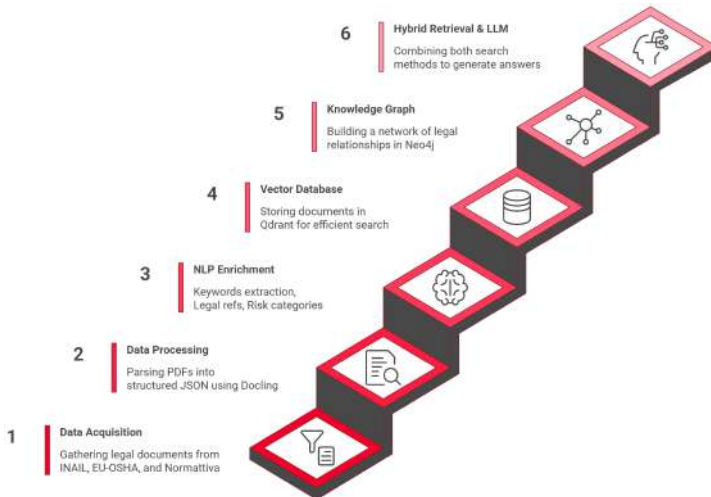


- **Deterministic + Probabilistic**
- Deterministic logic for consistent data integrity
- Verifiable retrieval focused on factual accuracy



GraphRAG Architecture: 6-Step Pipeline

1 Introduction



Key Terms

NLP

Processing and understanding natural language

Embeddings

Numeric vectors capturing semantic similarity

Knowledge Graph

Entities connected by typed relationships

RAG

Answer generation guided by retrieved information



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- ▶ Semantic Enrichment
- ▶ Vector Database Indexing
- ▶ Knowledge Graph Normative
- ▶ Hybrid GraphRAG



Corpus Construction: Sources and Scale

2 Data Acquisition

Sources

- **INAIL, EU-OSHA:** Institutional agencies
- **Normattiva:** Official legislative repository

Key Challenges

- No public APIs, JavaScript rendered content
- Heterogeneous layouts and metadata
- Large legal corpus requiring semantic filtering

Acquisition Results

Institutional Corpus (< 3% failures)

- INAIL: **771** parsed documents
- EU-OSHA: **1064** parsed documents

Normattiva Legal Corpus

- **100k+** acts (current legislation)
- **23k** OSH relevant laws after filtering



Web Scraping: From Websites to Structured Data

2 Data Acquisition

How it works

Tools

Automated browsers

HTML parsers

Output

Structured JSON files

Features

Respectful crawling

Source tracking

Extracted info

title, date, abstract

url, PDFs





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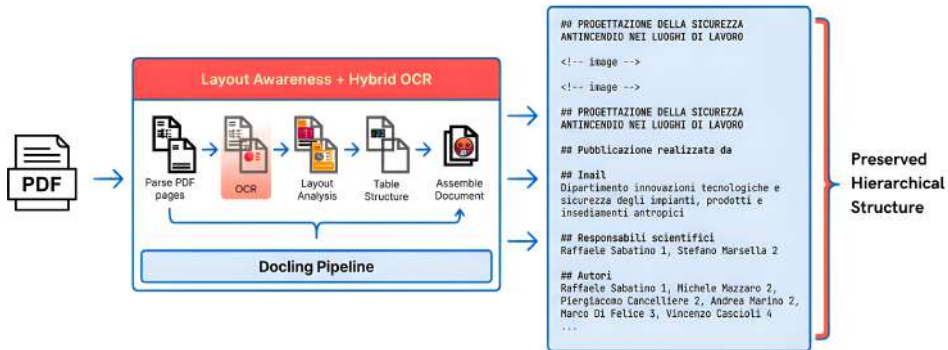
3 Document Parsing

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- ▶ **Document Parsing**
- ▶ Semantic Enrichment
- ▶ Vector Database Indexing
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Document Processing: From PDF to Structured Text

3 Document Parsing



Analyze

Recognize text, tables, and
page layout

Structure

Preserve document hierarchy
and formatting

Export

Output as Markdown or JSON
for further use



Docling: Tables & Figures to Searchable Text

3 Document Parsing

Standard extraction leaves
tables and figures **unsearchable**

Docling Solution:

Tables → Structured text

Preserving rows and columns

Figures → Text descriptions VLM
generated captions

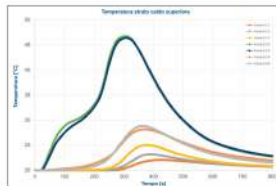
Fully searchable and retrievable

Input PDF

Classi di rischio	Descrizione	δ_{q2}
I	Area che presentano un basso rischio d'incendio in termini di probabilità d'innescio, velocità di propagazione delle fiamme e possibilità di controllo dell'incendio da parte delle squadre d'emergenza	0,00
II	Area che presentano un modesto rischio d'incendio in termini di probabilità d'innescio, velocità di propagazione delle fiamme e possibilità di controllo dell'incendio da parte delle squadre d'emergenza	1,00
III	Area che presentano un alto rischio d'incendio in termini di probabilità d'innescio, velocità di propagazione delle fiamme e possibilità di controllo dell'incendio da parte delle squadre d'emergenza	1,20

Output

```
| Classi di rischio | Descrizione | q2 |  
|-----|-----|-----|  
| I | Area che presentano un basso rischio d'incendio in termini di probabilità d'innescio, velocità di propagazione delle fiamme e possibilità di controllo dell'incendio da parte delle squadre d'emergenza | 0,00 |  
| II | Area che presentano un modesto rischio d'incendio in termini di probabilità d'innescio, velocità di propagazione delle fiamme e possibilità di controllo dell'incendio da parte delle squadre d'emergenza | 1,00 |  
| III | Area che presentano un alto rischio d'incendio in termini di probabilità d'innescio, velocità di propagazione delle fiamme e possibilità di controllo dell'incendio da parte delle squadre d'emergenza | 1,20 |
```



Main Content:

- The figure is a line chart plotting temperature (°C) versus time (s).
- Several curves rise from a baseline level, reach a peak, and then gradually decrease.
- The x-axis shows time (seconds) and the y-axis shows temperature (°C).
- Multiple curves peak at 300-350 s then decay toward different end points.
- Series start from about 20°C, peak around 300-350 s, and then gradually toward 23-25°C.



Parsing Results: The Scale of Extraction

3 Document Parsing



~100k

Total pages
(69.7k INAIL + 30k OSHA)

19k+

Table Extracted
(> 85% Success Rate)

35k+

Image description
(Fidelity Check Passed)

Processing Overview

- **Storage:** 67 GB PDF → 12 GB JSON
- **Compute:** 109 HPC Hours (Total)
- **Metadata:** Documents, Tables, Figures

Extraction Accuracy

Field	Precision
Document Title	96.5%
Reference ID	94.2%
Pub. Date	93.8%

Validation on 50 sample technical documents



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- ▶ Hybrid GraphRAG



Semantic Enrichment: A Verifiable Hybrid Approach

4 Semantic Enrichment

Problem: Pure LLMs

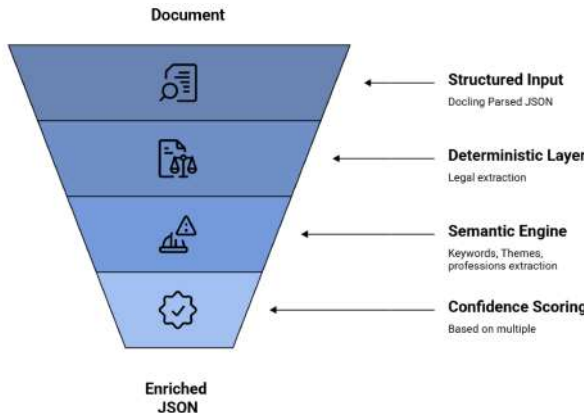
- Hallucinated legal references
- Inconsistent terminology
- No explainability

Solution: Hybrid Engine

Deterministic Patterns + Official Glossaries

Semantic Embeddings + TF-IDF

Reliability Multi-component Scoring





Legal Reference Extraction: Regex Based Engine

4 Semantic Enrichment

Multi-Type Extraction

- **Italy:** D.Lgs, DPR, Legge, DM, DL
- **EU:** Directives, Regulations

Granular Hierarchy

Law → *Article* → *Comma* → *Letter*

Contextual Windowing

±100 char. to capture Art. and Comma

Extraction	INAIL	EU-OSHA
Docs with ≥ 1 ref	94.3%	67.8%
Mean refs/doc	15.2	3.4

Law Identifier Example

```
D\.Lgs\.?\s*(?:n\.\s*)?  
(\d{1,4})[/\s-]+(\d{2,4})  
art\.?\s*(\d+)(?:,\s*\s*comma\s*(\d+))?
```

Captures:

D.Lgs. 81/2008, art. 29, comma 3

Normalization

- **Law ID:** dlgs-81-2008
- **Full citation:** dlgs-81-2008-art29-comma3
- Temporal validation (1900-2025)



Hybrid Keyword Extraction: Statistical + Semantic

4 Semantic Enrichment

Two Stage Pipeline

1. Statistical Filtering

TF-IDF + Domain Boosting

→ 30 candidates



$$TF_{t,d} = \frac{\text{count}(t,d)}{\sum \text{count}} \quad IDF_t = \log \frac{|D|}{|\{d : t \in d\}|}$$

$$\text{score}_{\text{boost}} = \begin{cases} 1.5 \times \text{TF-IDF} & t \in \text{glossary} \\ \text{TF-IDF} & \text{otherwise} \end{cases}$$

2. Semantic Ranking

EmbeddingGemma + Graph

→ 8 diverse keywords

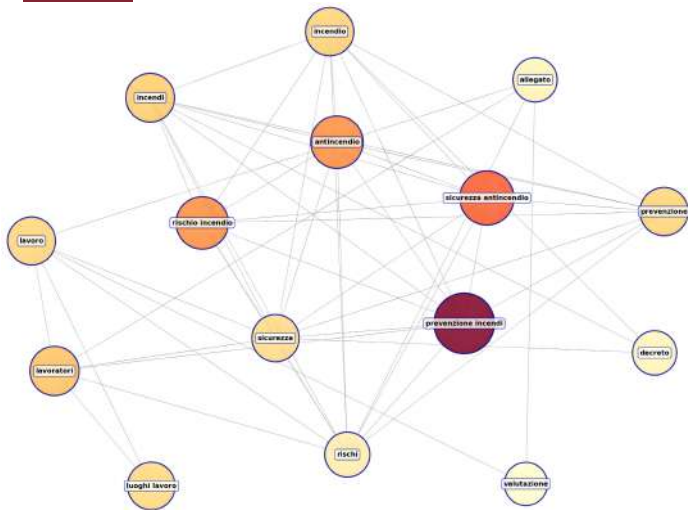


- **Embeddings:** Gemma-300M (384-dim, local)
- **Edges:** Link keywords with similarity > 0.7
- **Ranking:** PageRank ($\alpha = 0.85$) + Document similarity
- **Diversity:** Remove near duplicates if similarity > 0.85



Semantic Network & Graph-Based Ranking

4 Semantic Enrichment



Graph Construction

- **Nodes:** TF-IDF keyword candidates
- **Edges:** semantic similarity ($\cos > 0.7$)
- **Weights:** $w_{ij} = \cos(\mathbf{e}_i, \mathbf{e}_j)$

$$S_{\text{rank}} = 0.6 \cdot \text{PageRank} + 0.4 \cdot \text{DocSim}$$

- PageRank = centrality in semantic network
- DocSim = similarity to document embedding



Embedding Driven Semantic Categories

4 Semantic Enrichment



Thematic Clustering

Group similar keywords
into themes using embeddings
Automatic cluster detection

$$k = \min\left(4, \max\left(2, \left\lfloor \frac{|kw|}{3} \right\rfloor\right)\right)$$

Cosine distance, avg. linkage

Output:

Safety & Fire Prevention
Training & Evaluation



Risk Classification

Identify safety risks
combining term frequency
and semantic similarity

$$S_{final}(c) = 0.70 S_{term} + 0.30 S_{sem}$$

Explicit + implicit signals

Output:

Fire Risk (0.44)
Fall Protection (0.28)



Profile Mapping

Match against known roles
Filter out similar profiles
Use category as backup

Match: $\text{sim} > 0.60$

Diversity: $\text{sim} < 0.70$

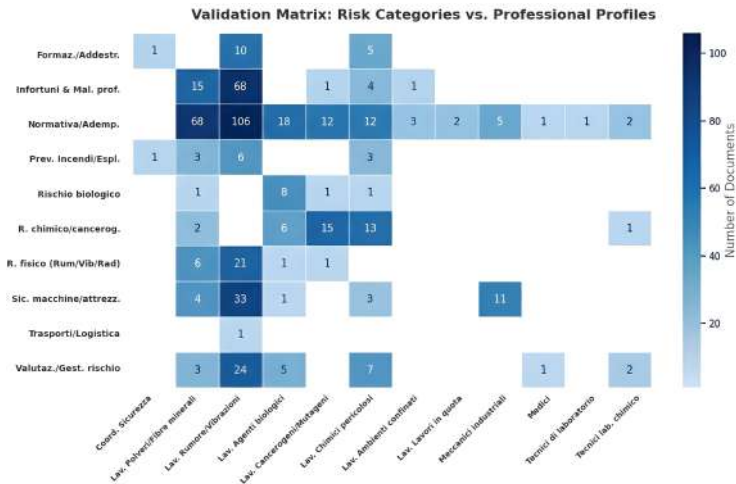
Output:

Safety Coordinators
Site Managers



Validation: Risk Profile Occurrence Matrix

4 Semantic Enrichment



Key Findings:

- Logical risk mapping (e.g., Chemical → Carcinogen)
- 68% sparsity: few false matches
- Risk categories cover multiple professions
- Embedding driven discovery, zero manual rules



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Vector Database: Semantic Retrieval Foundation

5 Vector Database Indexing

"Relational DBs store facts. Vector DBs store meaning."

- **Lexical gap:** keywords fail on domain language
("DPI" $\neq$ "PPE")
- **Solution:** move from **fields** to **meaning**

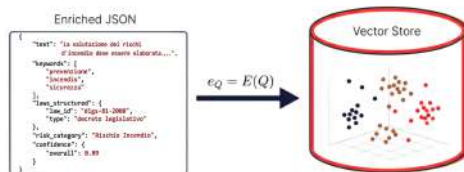
Semantic Matching

$$\text{sim}(Q, D) = \frac{\mathbf{e}_Q \cdot \mathbf{e}_D}{\|\mathbf{e}_Q\| \|\mathbf{e}_D\|}$$

Similarity = geometric proximity in $\mathbb{R}^{1024}$

Execution Layer

- **Qdrant + HNSW:** fast approximate search, $O(\log N)$
- **Metadata filters:** hybrid search capability



Key Optimizations

1. L2 Normalization

$$\|\mathbf{v}\| = 1 \Rightarrow \text{sim} = \mathbf{v}_i \cdot \mathbf{v}_j$$

Enables fast dot product search

2. Filtered Top-K Retrieval

$$\mathcal{R}_Q = \text{TopK}\{\text{sim}(\mathbf{q}, \mathbf{p}_i) \mid p_i \models F\}$$

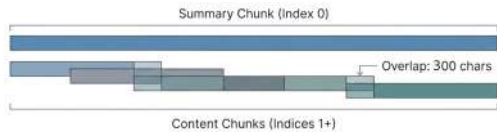
Metadata filters prune before ANN



Chunking Strategy & Retrieval Architecture

5 Vector Database Indexing

Natural Boundary Chunking



Hierarchy: Paragraphs → Sentences → Words → Hard split

Chunk size 4000 chars
Overlap 300 chars (7.5%)
Corpus size 57,328 chunks

Structure: $\left\{ \begin{array}{ll} \text{Index 0:} & \text{document summary + metadata} \\ \text{Index 1+:} & \text{content chunks with overlap} \end{array} \right.$

Legal Dual Granularity

From 23,759 Normattiva XMLs:

Collection	Level	Count
legal_context	Law	23,534
legal_articles	Article	186,286

Retrieval Capabilities

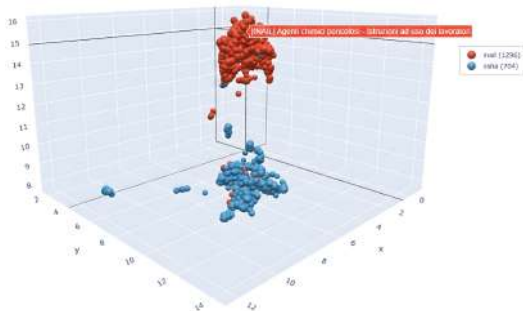
- **Semantic:** Cross-lingual (IT → EN)
- **Filtered:** By source, law, risk category
- **Multimodal:** Images + tables (38k vectors)
- **Precision:** Navigate from Law → Article
"Art. 29, comma 3, D.Lgs. 81/2008"



Embedding Validation: UMAP 3D Projection

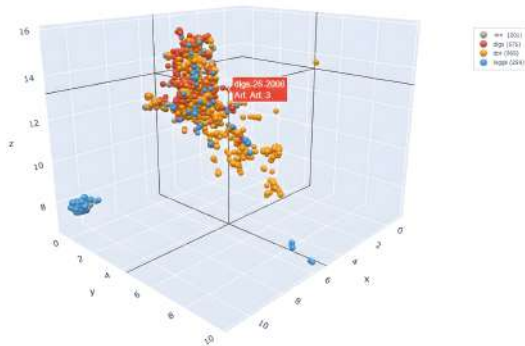
5 Vector Database Indexing

UMAP dimensionality reduction (1024-dim $\rightarrow$ 3D) preserves local structure while revealing global patterns



Drivers of separation

Safety topics • Regulatory approach
(IT vs EN/EU) • Documentation structure



Observed clustering

Topic based organization • Mixed legal forms
(D.Lgs, DPR, DM, Legge) • Thematic outliers



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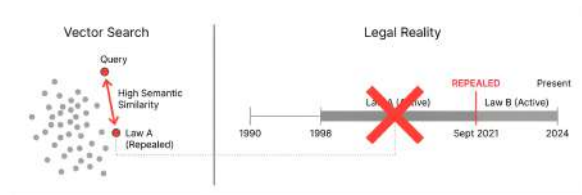
6 Knowledge Graph Normative

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From Flat RAG to GraphRAG: Validation Challenge

6 Knowledge Graph Normative



Vector finds context → **Graph** guarantees validity

Problem

- **Semantic $\neq$ Valid:** Similar text may be outdated or repealed
- **Temporal Risk:** Standard RAG ignores law evolution (vigenza)

Solution

- **Normative Graph:** Each node = a law
- **Relationships:** RICHIAMA, ABROGA, MODIFICA
- **Multi-layer:** Deterministic construction
- **Alignment:** law_id bridges Vector ↔ Graph



Knowledge Graph: Multi-Layer Construction Pipeline

6 Knowledge Graph Normative



Layer 0

Base Legal Entities

- Schema foundation:
- UNIQUE constraints
 - Temporal indexes
 - Vector→Graph bridge

186k articles from Qdrant

Output:

Law & Article nodes
with `law_id` alignment



Layer 1

Document Citations

- CITES relationships:
- 1,800 TechnicalDocs
 - `chunk_index` context
 - `citation_context`

Doc→Law connections

Output:

TechnicalDoc→Law
with positional context



Layer 2

Legal Relations

- Hybrid extraction:
- AKN XML `<ref>` tags
 - 20+ legal regex
 - LLM assisted

Temporal tracking

Output:

ABROGA, MODIFICA
SOSTITUISCE relations



Layer 3

Semantic Bridge

- Entity reuse:
- RiskCategory
 - ProfessionalTag
 - Topic

Risk/Role→Law bridge

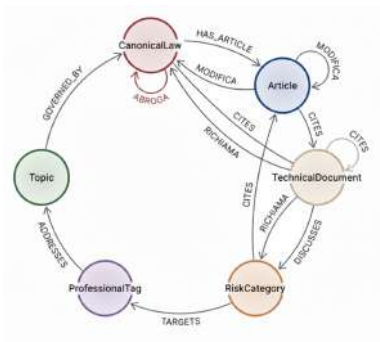
Output:

Semantic search
"Laws on fall protection"



Unified Knowledge Graph: Final Ontology

6 Knowledge Graph Normative



Network Scale: 6 Node types, 10 Relationship types

Architecture Design

- **Legal Backbone:** Law → Article → Paragraph with validity status
- **Semantic Shell:** Professional roles and risks categories linked to laws

Reasoning Path (Example)

1. Entry Point (Vector):

Query on "Asbestos handling in construction"

2. Semantic Anchor (Graph):

Match: RiskCategory:Asbestos

3. Deterministic Jump:

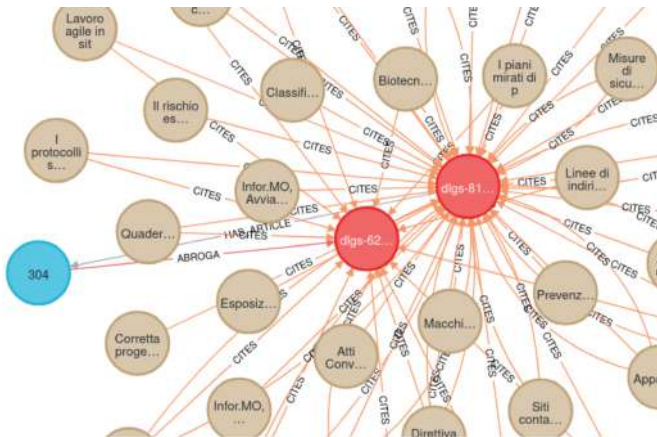
GOVERNED_BY → D.Lgs. 81/08

Outcome: Answers are not just "similar", they are legally validated and context aware



Temporal Validation: D.Lgs. 626/1994 Abrogation

6 Knowledge Graph Normative



Subgraph: 50 technical documents citing D.Lgs. 626/1994

Case Study:

D.Lgs. 626/1994
(old OSH main law)

ABROGATO DA

↓ Article 304

↓ D.Lgs. 81/2008
(current legislation)

Graph Enables:

- Trace links to related documents
- Current law detection
- Track validity over time



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- ▶ **Hybrid GraphRAG**



GraphRAG: Hybrid Retrieval in Action

7 Hybrid GraphRAG

```
QUERY: sicurezza antincendio

#1 | Score: 0.9163 (vector: 0.8726)
  | Collection: technical_documents_text
  | Progettazione della sicurezza antincendio nei luoghi di lavoro
  | Source: INAIL
  | Categoria: Prevenzione incendi ed esplosioni
  | Laws (9): dpr-175-1988, dlgs-81-2008, dpr-547-1955, dm-01-09-2021,
  | dpr-547-1955 ABROGATA da: dlgs-81-2008
  | dlgs-626-1994 ABROGATA da: dlgs-81-2008
  | Graph: 7 vigenti, 1 modificate, 2 abrogate
  | # Progettazione della sicurezza antincendio nei luoghi di lavoro

#2 | Score: 0.8965 (vector: 0.8747)
  | Collection: technical_documents_text
  | La protezione attiva antincendio
  | Source: INAIL
  | Categoria: Altro
  | Laws (1): dm-20-12-2012
  | Graph: 1 vigenti, 0 modificate, 0 abrogate
  | | 284 | | Inquadramento nel Codice

#3 | Score: 0.8834 (vector: 0.8671)
  | Collection: technical_documents_text
  | Rischio incendio ed esplosione in agricoltura
  | Source: INAIL
  | Categoria: Prevenzione incendi ed esplosioni
  | Laws (25): dlgs-139-2006, dlgs-81-2008, dm-8-11-2019, dlgs-105-2015, dm
  | Graph: 25 vigenti, 0 modificate, 0 abrogate
  | prevenzione incendi. Progettare la sicurezza antincendio di un'att
```

Query: "sicurezza antincendio"

Pipeline:

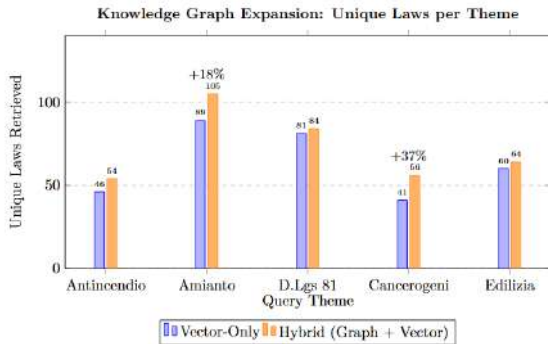
1. **Vector Search:** Semantic matching across 5 collections
2. **Graph Validation:** KG checks ABROGA/MODIFICA relations
3. **Safety Warnings:** Flags outdated laws (DPR 547-1955, DLGS 626-1994)

Result: 7 valid + 2 repealed in 2.8s



Evaluation: Graph Expansion & Legal Validation

7 Hybrid GraphRAG



Knowledge Graph law expansion across 10 query per themes

Legal Validation Results

Metric	Vector	Hybrid
Abrogation Recall	0%	100%
Queries w/ warnings	0	42%
Avg. latency	1.1s	3.8s



Trade off

+2.7s latency for legal accuracy

Acceptable for compliance

tasks



Key Insight

Temporal Awareness via
Graph layer

Zero semantic quality loss



LLM Integration: Qwen 2.5 72B

7 Hybrid GraphRAG

RAG Pipeline

1. Query Classification

Legal / Procedural / General

2. Hybrid Retrieval

Qdrant (semantic) + Neo4j (validation)

3. Context Construction

Warnings → Documents + Graph enrichment

4. Generation

Qwen 2.5 72B · Temp: 0.15 · Max: 1500 tokens

 **Live Demo:**
gradio.live demo

Example Response:

Query: "Leggi relative a procedure sicurezza antincendio"

Domanda Giuridica | 61.1s

Attenzione: Nei documenti consultati sono presenti riferimenti a **4 leggi ABROGATE**. La risposta tiene conto di questo aspetto.

Risposta

La valutazione del rischio incendio in ambito lavorativo è regolamentata da diverse normative. Ecco un riepilogo delle principali disposizioni:

1. D.Lgs. 81/2008 (Testo Unico sulla Sicurezza e la Salute sul Lavoro):

- **Articolo 9:** Impone l'obbligo di valutazione dei rischi per i lavoratori, compresi quelli legati all'incendio.
- **Articolo 20:** Stabilisce che il datore di lavoro deve predisporre un piano di emergenza in caso di incendio e altre situazioni d'emergenza.
- **Allegato I (Sezione II, Punto 1.3):** Indica specificamente la necessità di valutare i rischi legati all'incendio.

2. D.M. 10 marzo 1998:

- **All. IX:** Classifica i luoghi di lavoro in base al rischio d'incendio, suddividendoli in tre categorie: basso, medio e elevato.
- **Punto 9.2:** Definisce le attività soggette a classificazione come luoghi a rischio d'incendio elevato.
- **Punto 9.3:** Definisce le attività soggette a classificazione come luoghi a rischio d'incendio medio.

3. D.Lgs. 28 aprile 1995, n. 230:

- **Articolo 7:** Impone l'obbligo di adottare misure preventive e protettive per prevenire incidenti sul lavoro, compresi quelli legati all'incendio.
- **Allegato I (Sezione II):** Indica le procedure per la valutazione dei rischi, inclusi quelli d'incendio.

4. D.M. 20 dicembre 1982:

- **Articolo 3:** Stabilisce norme tecniche di sicurezza e igiene del lavoro in ambienti a rischio d'incendio.
- **Allegato A (Sezione II):** Fornisce dettagli specifici sulla valutazione dei rischi legati all'incendio.

5. Legge 30 aprile 2007, n. 81:

- **Articolo 1:** Introduce il D.Lgs. 81/2008 e ne stabilisce gli obiettivi generali.
- **Articolo 46:** Regola la classificazione dei luoghi di lavoro in base al rischio d'incendio, riferendosi alle disposizioni del D.M. 10 marzo 1998.



Conclusions & Future Work

7 Hybrid GraphRAG



Key Contributions

Validity-Aware Retrieval

Temporal legal awareness (vigente / abrogata)

Hybrid GraphRAG

Vector search + legal graph validation

Large-Scale OSH Corpus

2k+ documents, 186k articles, fully searchable



Future Directions

EU-Wide Coverage

Extend graph to all EU Directives

Domain Specific LLM

Fine tuning on OSH terminology

Live Legal Monitoring

Auto-update graph when laws change



Design, Development and Experimentation of an LLM Agent for the Workplace Safety

Thank you for listening!
Any questions?